

Body Control Module (BCM)



Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Body Control Module (BCM) controls various systems by monitoring inputs from switches, sensors and messages sent from other modules. The BCM activates outputs based on the inputs received. For example, the Steering Column Control Module (SCCM) monitors the headlamp switch and transmits the headlamp switch position to the BCM over the network. Based on this input, the BCM may provide voltage to the exterior lamps.

Field-Effect Transistor (FET) Protection

A Field-Effect Transistor (FET) is a type of transistor that, when used with module software, monitors and controls current flow on module outputs. The FET protection strategy prevents module damage in the event of excessive current flow.

The BCM utilizes an FET protective circuit strategy for many of its outputs (for example, a headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault event is detected. A short circuit DTC is stored at the fault event and a cumulative counter is started.

When the demand for the output is no longer present, the module resets the FET protection to allow the circuit to function. The next time the driver requests a circuit to activate that has been shut down by a previous short (FET protection) and the circuit is still shorted, the FET protection shuts off the circuit again and the cumulative counter advances.

When the excessive circuit load occurs often enough, the module shuts down the output until a repair procedure is carried out. Each FET protected circuit has 3 predefined levels of short circuit tolerance based on the harmful effect of each circuit fault on the FET and the ability of the FET to withstand it. A module lifetime level of fault events is established based upon the durability of the FET . If the total tolerance level is determined to be 600 fault events, the 3 predefined levels would be 200, 400 and 600 fault events.

When each tolerance level is reached, DTC U1000:00 should set along with the short circuit DTC that was stored on the first failure. These DTCs cannot be cleared until the vehicle is repaired.

After the repair, it is necessary for the module successfully complete a self-test (the module no longer detects the fault), to clear the DTCs using the Clear DTC operation on the scan tool, and then cycling the ignition for the circuit function to return.

The module never resets the fault event counter to zero and continues to advance the fault event counter as short circuit fault events occur. If the number of short circuit fault events reach the third level, then DTCs U1000:00 and U3000:49 set along with the associated short circuit DTC. DTC U3000:49 cannot be cleared and the module must be replaced after the initial fault is repaired.

Gateway Function

The BCM acts as a gateway module by receiving information in one format and transmitting it to other modules using another format. For example, the BCM receives the vehicle speed data from the PCM over the High Speed Controller Area Network (HS-CAN) , converts the data into a Medium Speed Controller Area Network (MS-CAN) message and sends (gateways) the message to other network modules such as the HVAC module and the Audio Front Control Module (ACM) . This enables network communication between modules that do not communicate using the same network (HS-CAN or MS-CAN).

Inspection and Verification

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical
• Body Control Module (BCM)

- 3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
- 4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

- 5. **NOTE:** *The Vehicle Communication Module (VCM) prove-out confirms power and ground from the DLC are provided to the VCM .*

- If the scan tool does not communicate with the VCM :
- check the VCM connection to the vehicle.
 - check the scan tool connection to the VCM .
 - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

- 6. If the scan tool does not communicate with the vehicle:
 - verify the ignition is ON.
 - The air bag warning indicator prove-out confirms ignition ON (other indicators may not prove ignition ON). If the ignition does not turn on, refer to Section 211-05 to diagnose no power in RUN.
 - verify the scan tool operation with a known good vehicle.

- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication with one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the **BCM** .

9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B1048:01	Brake Fluid Level Switch: General Electrical Failure	REFER to Section 413-01.
B10AB:00	Remote Keyless Entry Synchronization: No Sub Type Information	REFER to Section 501-14.
B10D5:13	PATS Antenna: Circuit Open	REFER to Section 419-01B.
B10D7:05	PATS Key: System Programming Failure	REFER to Section 419-01B.
B10D7:51	PATS Key: Not Programmed	REFER to Section 419-01B.
B10D7:55	PATS Key: Not Configured	REFER to Section 419-01B.
B10D7:87	PATS Key: Missing Message	REFER to Section 419-01B.
B10D7:8F	PATS Key: Erratic	REFER to Section 419-01B.
B10D8:00	PATS Key Less Than Minimum Programmed: No Sub Type Information	REFER to Section 419-01B.
B10D9:87	PATS Transponder: Missing Message	REFER to Section 419-01B.
B10DA:51	PATS Target Identifier: Not Programmed	REFER to Section 419-01B.
B10F1:14	Key In switch: Circuit Short To Ground or Open	REFER to Section 413-01.
B1115:11	High Mounted Stop Lamp control: Circuit Short To Ground	REFER to Section 417-01.
B1142:29	Ignition Status 1: Signal Invalid	REFER to Section 211-05.
B1175:01	Drivers Door Ajar Switch: General Electrical Failure	REFER to Section 417-02.
B1176:01	Passenger Door Ajar Switch: General Electrical Failure	REFER to Section 417-02.

DTC	Description	Action
B1182:00	Tire Pressure Monitoring System: No Sub Type Information	REFER to Section 204-04.
B1182:55	Tire Pressure Monitoring System: Not Configured	REFER to Section 204-04.
B11C0:01	Driver Side Rear Door Ajar Switch: General Electrical Failure	REFER to Section 417-02.
B11C1:01	Passenger Side Rear Door Ajar Switch: General Electrical Failure	REFER to Section 417-02.
B1218:00	Transmitter Identification Code: No Sub Type Information	REFER to Section 501-14.
B121A:11:	Keypad Illumination Output: Circuit Short To Ground	REFER to Section 501-14.
B121B:01	Keypad Input Switch: General Electrical Failure	REFER to Section 501-14.
B123A:11	Left Front Turn Indicator: Circuit Short To Ground	REFER to Section 417-01.
B123A:15	Left Front Turn Indicator: Circuit Short To Battery or Open	REFER to Section 417-01.
B123B:11	Right Front Turn Indicator: Circuit Short To Ground	REFER to Section 417-01.
B123B:15	Right Front Turn Indicator: Circuit Short To Battery or Open	REFER to Section 417-01.
B124D:02	Tire Pressure Sensor: General Signal Failure	REFER to Section 204-04.
B124D:4A	Tire Pressure Sensor: Incorrect Component Installed	REFER to Section 204-04.
B1250:00	Tire Pressure Monitoring System (TPMS) Sensor Pressure Range: No Sub Type Information	REFER to Section 204-04.
B1251:00	Tire Pressure Sensor Low Battery: No Sub Type Information	REFER to Section 204-04.
B1254:51	Right Rear (Outside on Duel Wheel) Tire Pressure Sensor and Transmitter Assembly: Not Programmed	REFER to Section 204-04.
B1255:51	Left Rear (Outside On Duel Wheel) Tire Pressure Sensor and Transmitter Assembly: Not Programmed	REFER to Section 204-04.

DTC	Description	Action
B126A:01	Drivers Door Interior Lock Switch: General Electrical Failure	REFER to Section 501-14.
B1277:11	Reverse Lamp: Circuit Short To Ground	REFER to Section 417-01.
B1277:15	Reverse Lamp: Circuit Short To Battery or Open	REFER to Section 417-01.
B12C2:11	Puddle Lamp: Circuit Short To Ground	REFER to Section 417-02.
B1305:01	Hood Switch: General Electrical Failure	REFER to Section 419-01A.
B130C:12	Load Shed Control: Circuit Short To Battery	REFER to Section 417-01.
B130C:14	Load Shed Control: Circuit Short To Ground or Open	REFER to Section 417-01.
B130D:00	Memory Full for Integrated Key Programming: No Sub Type Information	REFER to Section 501-14.
B1310:12	Run/Start Control: Circuit Short To Battery	REFER to Section 211-05.
B1310:14	Run/Start Control: Circuit Short To Ground or Open	REFER to Section 211-05.
B1313:11	Interior Lighting Output: Circuit Short To Ground	REFER to Section 417-02.
B1313:15	Interior Lighting Output: Circuit Short To Battery or Open	REFER to Section 417-02.
B1315:11	Backlighting (Non Reflective Controls) Illumination Output: Circuit Short To Ground	REFER to Section 413-00.
B1319:11	Shift Interlock (BSI) Output Circuit: Circuit Short To Ground	REFER to Section 307-05.
B1319:15	Shift Interlock (BSI) Output Circuit: Circuit Short To Battery or Open	REFER to Section 307-05.
B131A:12	Delayed Accessory Power: Circuit Short To Battery	REFER to Section 501-11.
B131A:14	Delayed Accessory Power: Circuit Short To Ground or Open	REFER to Section 501-11.
B132B:11	Left Stop/Turn Lamp: Circuit Short To Ground	REFER to Section 417-01.
B132B:15	Left Stop/Turn Lamp: Circuit Short To Battery or Open	REFER to Section 417-01.
B132C:11	Right Stop/Turn Lamp: Circuit Short To Ground	REFER to Section 417-01.

DTC	Description	Action
B132C:15	Right Stop/Turn Lamp: Circuit Short To Battery or Open	REFER to Section 417-01.
B1331:02	Compass/Mirror Module: General Signal Failure	- If the vehicle is equipped with an auto-dimming interior mirror, REFER to Section 413-13. - If the vehicle is not equipped with an auto-dimming interior mirror, REFER to Section 415-00B.
B1331:08	Compass/Mirror Module: Bus Signal/Message Failure	- If the vehicle is equipped with an auto-dimming interior mirror, REFER to Section 413-13. - If the vehicle is not equipped with an auto-dimming interior mirror, REFER to Section 415-00B.
B1331:49	Compass/Mirror Module: Internal Electronic Failure	- If the vehicle is equipped with an auto-dimming interior mirror, REFER to Section 413-13. - If the vehicle is not equipped with an auto-dimming interior mirror, REFER to Section 415-00B.
B1A85:11	Ambient Light Sensor: Circuit Short To Ground	REFER to Section 417-01.
B1A85:13	Ambient Light Sensor: Circuit Open	REFER to Section 417-01.
B1C98:11	Left Corner Lamp Circuit: Circuit Short To Ground	REFER to Section 417-01.
B1C99:11	Right Corner Lamp Circuit: Circuit Short To Ground	REFER to Section 417-01.
B1D00:11	Left Low Beam: Circuit Short To Ground	REFER to Section 417-01.
B1D01:11	Right Low Beam: Circuit Short To Ground	REFER to Section 417-01.
B1D02:11	Left High Beam Circuit: Circuit Short To Ground	REFER to Section 417-01.
B1D02:15	Left High Beam Circuit: Circuit Short To Battery or Open	REFER to Section 417-01.
B1D03:11	Right High Beam Circuit: Circuit Short To Ground	REFER to Section 417-01.
B1D03:15	Right High Beam Circuit: Circuit Short To Battery or Open	REFER to Section 417-01.
B1D35:11	Hazard Switch: Circuit Short To Ground	REFER to Section 417-01.
C0040:12	Brake Pedal Switch "A": Circuit Short To Battery	REFER to Section 417-01.
C113A:11	Wakeup Control: Circuit Short To Ground	REFER to Section 419-01B (Passive Anti-Theft System (PATS)).
C113A:15	Wakeup Control: Circuit Short To Battery or Open	REFER to Section 419-01B (PATS)).

DTC	Description	Action
C1A56:51	Left Front Tire Pressure Sensor and Transmitter Assembly: Not Programmed	REFER to Section 204-04.
C1A58:51	Right Front Tire Pressure Sensor and Transmitter Assembly: Not Programmed	REFER to Section 204-04.
C1D00:11	Park Brake Apply Switch: Circuit Short To Ground	REFER to Section 413-01.
P0230:12	Fuel Pump Primary Circuit: Circuit Short To Battery	DIAGNOSE PCM DTC P025A. REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.
P0230:14	Fuel Pump Primary Circuit: Circuit Short To Ground or Open	DIAGNOSE PCM DTC P025A. REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.
U0100:00	Lost Communication With ECM/PCM "A": No Sub Type Information	GO to Pinpoint Test A.
U0121:00	Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub Type Information	GO to Pinpoint Test B.
U0131:00	Lost Communication With Power Steering Control Module: No Sub Type Information	GO to Pinpoint Test C.
U0182:00	Lost Communication With Lighting Control Module - Front: No Sub Type Information	GO to Pinpoint Test D.
U0212:00	Lost Communication With Steering Column Control Module: No Sub Type Information	GO to Pinpoint Test E.
U1000:00	Solid State Driver Protection Activated - Driver Disabled: No Sub Type Information	GO to Pinpoint Test F.
U1007:31	Lost Communication With Battery Monitoring Sensor "A": No Signal	REFER to Section 414-00.
U2005:85	Vehicle Speed: Signal Above Allowable Range	A speed signal is detected during the self-test. DIAGNOSE any symptoms or other module DTCs that are present. CLEAR the DTCs. MAKE SURE the vehicle is not moving and REPEAT the Body Control Module (BCM) self-test. If no other symptoms or DTCs are present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CARRY OUT Programmable Module Installation (PMI) on the BCM . REFER to Section 418-01. CLEAR the DTCs. REPEAT the self-test. If DTC U2100:00 is retrieved again, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

DTC	Description	Action
U2200:55	Control Module Configuration Memory Corrupt: Not Configured	CARRY OUT PMI on the BCM . REFER to Section 418-01. CLEAR the DTCs. REPEAT the self-test. If DTC U2200:55 is retrieved again, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
U3000:49	Control Module: Internal Electronic Failure	ADDRESS all other DTCs first. CLEAR the DTCs. REPEAT the self-test. If DTC U3000:49 is retrieved again, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test G.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test H.
U300A:01	Ignition Switch: General Electrical Failure	REFER to Section 211-05.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
No communication with the Body Control Module (BCM)	- Wiring, terminals or connectors - BCM	REFER to Section 418-00, The BCM Does Not Respond To The Scan Tool.

Pinpoint Tests

Pinpoint Test A: DTC U0100:00

Normal Operation

- DTC U0100:00 (Lost Communication With ECM/PCM "A": No Sub Type Information) — sets in the Body Control Module (BCM) if data messages received (vehicle speed) from the PCM are missing for more than 1 second.

This pinpoint test is intended to diagnose the following:

- Communication network concern
- PCM
- BCM

PINPOINT TEST A : DTC U0100:00

A1 VERIFY THE SCAN TOOL CAN COMMUNICATE WITH THE PCM	
- Connect the scan tool. - Check that a vehicle session can be established using the scan tool.	
Can a vehicle session be established?	
Yes	GO to A2 .
No	REFER to Section 418-00, The PCM Does Not Respond To The Scan Tool.

A2 CHECK THE BCM CONTINUOUS MEMORY DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **BCM Self-Test** .
- Clear the DTCs. Repeat the **BCM** self-test.

Is DTC U0100:00 retrieved again?

Yes	GO to A3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

A3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded **BCM** DTCs from the self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test G . For DTC U3003:17, GO to Pinpoint Test H .
No	GO to A4 .

A4 CHECK FOR DTC U0100, U0100:00 OR U0100:87 SET IN OTHER MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0100, U0100:00 or U0100:87 set in any other module?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test B: DTC U0121:00

Normal Operation

- DTC U0121:00 (Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub Type Information) — sets in the Body Control Module (BCM) if data messages received from the ABS module are missing.

This pinpoint test is intended to diagnose the following:

- Communication network concern
- ABS module
- BCM

PINPOINT TEST B : DTC U0121:00

B1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to B2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

B2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the ABS module pass the network test?

Yes	GO to B3 .
No	REFER to Section 418-00, The ABS Module Does Not Respond To The Scan Tool.

B3 RETRIEVE THE RECORDED DTCS FROM THE BCM AND SELF-TEST

- Check for recorded ~~BCM~~ DTCs from the self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test G . For DTC U3003:17, GO to Pinpoint Test H .
No	GO to B4 .

B4 RECHECK THE BCM DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during Programmable Module Installation (PMI) or the ~~PMI~~ may not have been carried out.*

- Clear the DTCs. Repeat the ~~BCM~~ self-test.

Is DTC U0121:00 still present?

Yes	GO to B5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

B5 CHECK FOR DTC U0121, U0121:00 OR U0121:87 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0121, U0121:00 or U0121:87 set in any other module?

Yes	INSTALL a new ABS module. REFER to Section 206-09. CLEAR the DTCs. REPEAT the BCM self-test.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test C: DTC U0131:00

Normal Operation

- DTC U0131:00 (Lost Communication With Power Steering Control Module: No Sub Type Information) — sets in the Body Control Module (BCM) if data messages received from the Power Steering Control Module (PSCM) are missing.

This pinpoint test is intended to diagnose the following:

- Communication network concern
- PSCM
- BCM

PINPOINT TEST C : DTC U0131:00

C1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to C2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

C2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the PSCM pass the network test?

Yes	GO to C3 .
No	REFER to Section 418-00, The PSCM Does Not Respond To The Scan Tool.

C3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCs from the self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test G . For DTC U3003:17, GO to Pinpoint Test H .
No	GO to C4 .

C4 RECHECK THE BCM DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs. Repeat the BCM self-test.

Is DTC U0131:00 still present?

Yes	GO to C5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

C5 CHECK FOR DTC U0131, U0131:00 OR U0131:87 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0131, U0131:00 or U0131:87 set in any other module?

Yes	INSTALL a new Electronic Power Assist Steering (EPAS) steering gear. REFER to Section 211-02. CLEAR the DTCs. REPEAT the BCM self-test.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test D: DTC U0182:00

Normal Operation

- DTC U0182:00 (Lost Communication With Lighting Control Module - Front: No Sub Type Information) — sets in the Body Control Module (BCM) if headlamp switch data messages received from the Steering Column Control Module (SCCM) are missing.

This pinpoint test is intended to diagnose the following:

- Communication network concern
- SCCM
- BCM

PINPOINT TEST D : DTC U0182:00

D1 VERIFY THE CUSTOMER CONCERN	
• Ignition ON. • Verify that there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to D2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
D2 CHECK THE COMMUNICATION NETWORK	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Network Test . • Carry out the network test.	
Does the SCCM pass the network test?	
Yes	GO to D3 .
No	REFER to Section 418-00, The SCCM Does Not Respond To The Scan Tool.

D3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCs from the self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test G . For DTC U3003:17, GO to Pinpoint Test H .
No	GO to D4 .

D4 RECHECK THE BCM DTCS

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs. Repeat the BCM self-test.

Is DTC U0182:00 still present?

Yes	GO to D5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

D5 CHECK FOR DTC U0182, U0182:00 OR U0182:87 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0182, U0182:00 or U0182:87 set in any other module?

Yes	INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. REPEAT the BCM self-test.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test E: DTC U0212:00

Normal Operation

- DTC U0212:00 (Lost Communication With Steering Column Control Module: No Sub Type Information) — sets in the Body Control Module (BCM) if turn signal switch data messages received from the Steering Column Control Module (SCCM) are missing.

This pinpoint test is intended to diagnose the following:

- Communication network concern
- SCCM
- BCM

PINPOINT TEST E : DTC U0212:00

E1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to E2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

E2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

Does the SCCM pass the network test?

Yes	GO to E3 .
No	REFER to Section 418-00, The SCCM Does Not Respond To The Scan Tool.

E3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST

- Check for recorded BCM DTCs from the self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test G . For DTC U3003:17, GO to Pinpoint Test H .
No	GO to E4 .

E4 RECHECK THE BCM DTCS

NOTE: If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during Programmable Module Installation (PMI) or the PMI may not have been carried out.

- Clear the DTCs. Repeat the BCM self-test.

Is DTC U0212:00 still present?

Yes	GO to E5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

E5 CHECK FOR DTC U0212, U0212:00 OR U0212:87 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

Is DTC U0212, U0212:00 OR U0212:87 set in any other module?

Yes	INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. REPEAT the BCM self-test.
No	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

Pinpoint Test F: DTC U1000:00

Normal Operation

The Body Control Module (BCM) controls the output of several vehicle systems using solid state drivers. A DTC sets when an overload occurs on any of these drivers. The module tracks the number of repetitive faults on each of these circuits and compares this number of overloads to 3 progressive thresholds established for each circuit.

At the each threshold, DTC U1000:00 sets along with the DTC associated with the affected circuit.

- DTC U1000:00 (Solid State Driver Protection Activated - Driver Disabled: No Sub Type Information) — sets when the BCM has disabled a circuit due to a repetitive circuit overload and a progressive threshold is met.

This pinpoint test is intended to diagnose the following:

- Output circuit short
- BCM

PINPOINT TEST F : DTC U1000:00

F1 REVIEW THE DTCS

NOTICE: First follow the necessary diagnostics to resolve DTC U1000:00 and any other present DTCs (ignoring U3000:49). DTC U1000:00 is only present when there is another DTC due to software or circuitry

issues that must be resolved before another self-test is run. DTC U3000:49 is only present once the final threshold has been met. Failure to correct the fault condition first may cause damage to the new BCM , resulting in a repeat repair.

- Review the DTCs from the BCM self-test.

Are any other DTCs present, besides U1000:00 or U3000:49?

Yes	CORRECT the DTCs. REFER to the Body Control Module (BCM) DTC chart in this section. After the repair, GO to F2 .
No	GO to F2 .

F2 REPEAT THE BCM SELF-TEST

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM Self-Test .
- Clear the DTCs and repeat the BCM self-test.

Is DTC U3000:49 present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The circuit short has been repaired.

Pinpoint Test G: DTC U3003:16

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/SJB for schematic and connector information.

Normal Operation

- DTC U3003:16 (Battery Voltage: Circuit Voltage Below Threshold) — sets in the Body Control Module (BCM) when the BCM detects that battery voltage has dropped below 9 volts on the voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- High circuit resistance
- BCM

PINPOINT TEST G : DTC U3003:16

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Failure to disconnect the battery when instructed results in false resistance readings. Refer to Section 414-01.
G1 RECHECK THE BCM DTCS
• Ignition ON.

- Enter the following diagnostic mode on the scan tool: **BCM** Self-Test .
- Clear the DTCs. Repeat the **BCM** self-test.

Is DTC U3003:16 still present?

Yes	GO to G2 .
No	The system is operating normally at this time. The DTC may have been set due to a previous low battery voltage condition.

G2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM

- Enter the following diagnostic mode on the scan tool: PCM Self-Test .
- Carry out the PCM Key ON Engine OFF (KOEO) self-test.
- Retrieve the continuous memory DTCs from the PCM.

Is DTC P0620, P0622, P0625, P0626 or P065B set in the PCM?

Yes	REFER to Section 414-00.
No	GO to G3 .

G3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE

- Check the battery condition and verify the battery is fully charged. Refer to Section 414-01.

Is the battery OK and fully charged?

Yes	GO to G4 .
No	REFER to Section 414-00.

G4 CHECK THE BCM VOLTAGE PID

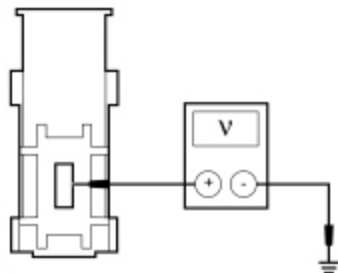
- Ignition ON.
- Measure and record the voltage at the battery.
- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Monitor the **BCM** voltage PID (V_BATT).

Is the voltage within 0.2 volt of the recorded battery voltage?

Yes	GO to G7 .
No	GO to G5 .

G5 CHECK THE BCM VOLTAGE SUPPLY

- Ignition OFF.
- Disconnect: BCM [C2280G](#) .
- Ignition ON.
- Measure the voltage between the BCM [C2280G](#) Pin 1, circuit SDC04 (RD), harness side and ground.



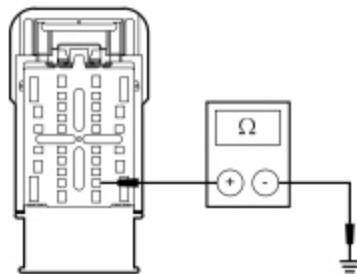
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Is the voltage within 0.2 volt of the recorded battery voltage?

Yes	GO to G6 .
No	REPAIR circuit SDC04 (RD) for high resistance. CLEAR the DTC. REPEAT the self-test.

G6 CHECK THE BCM GROUND CIRCUIT FOR HIGH RESISTANCE

- Ignition OFF.
- Disconnect: Negative Battery Cable .
- Disconnect: BCM [C2280D](#) .
- Measure the resistance between the BCM [C2280D](#) Pin 25, circuit GD139 (BK/YE), harness side and ground.



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Is the resistance less than 5 ohms?

Yes	GO to G7 .
No	REPAIR circuit GD139 (BK/YE) for high resistance. CLEAR the DTC. REPEAT the self-test.

G7 CHECK FOR CORRECT BCM OPERATION

- Disconnect the BCM connectors.

- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: DTC U3003:17

Refer to Wiring Diagrams Cell [13](#) , Power Distribution/SJB for schematic and connector information.

Normal Operation

- DTC U3003:17 (Battery Voltage: Circuit Voltage Above Threshold) — sets in the Body Control Module (BCM) when the **BCM** detects that battery voltage is more than 16 volts on the voltage supply circuit.

This pinpoint test is intended to diagnose the following:

- Charging system concern
- **BCM**

PINPOINT TEST H : DTC U3003:17

NOTE: <i>DTC U3003:17 may be stored in the module memory due to previous battery charging or vehicle jump starting events.</i>	
H1 CHECK FOR DTC B1317, B1676, P0563 (PCM) OR U3003:17 SET IN OTHER MODULES	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Self-Test . • Retrieve the continuous memory DTCs from all modules.	
Is DTC B1317, B1676, P0563 (PCM) or U3003:17 set in more than one module?	
Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to H2 .
H2 CHECK THE BATTERY VOLTAGE	
• Turn off all interior/exterior lights and accessories.	

- Start and run the engine at approximately 2,000 rpm for 3 minutes while monitoring the battery voltage.

Does the battery voltage rise to 15.5 volts or higher?

Yes	REFER to Section 414-00 to diagnose an overcharging condition.
No	GO to H3 .

H3 RECHECK FOR DTC U3003:17

- Turn the engine off.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **BCM Self-Test** .
- Clear the continuous memory DTCs.
- Carry out the **BCM** self-test.

Is DTC U3003:17 present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating normally at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle.